

NutriLabelAI: An AI-Powered Nutrition Label Generation System

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NutriLabelAI Team

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Submitted to:

Dr. Stefan Andrei

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Executive Summary

NutriLabelAI is a full-stack artificial intelligence system designed to generate FDA-style nutrition labels from user input, specifically dish names, calorie targets, or food images. This proposal outlines a comprehensive mobile and backend application that addresses the critical gap in nutrition data accessibility and accuracy for consumers eating at restaurants or cooking at home.

The core problem addressed is the inconsistency and fragmentation of public nutrition data, which makes it difficult for users to obtain accurate macronutrient breakdowns for non-packaged foods. Existing solutions often require tedious manual ingredient entry or rely on generic databases that do not account for preparation variations.

NutriLabelAI solves this by leveraging an ML ensemble methodology. The system utilizes embedding-based retrieval (Sentence-BERT) to identify representative dish profiles, followed by a regression model and neural network refinement to scale nutrients to specific calorie targets. The deliverables include a React Native mobile application for iOS and Android, a high-performance FastAPI backend, and a production-ready PostgreSQL database with vector search capabilities, delivering fast, consistent, and explainable nutrition estimates to the user.

1. Problem Statement and Background

Consumers face significant challenges when attempting to obtain accurate nutritional information for meals, whether prepared at home or purchased from restaurants. The fundamental issue is data fragmentation and ambiguity. For example, a dish like "Butter Chicken" varies dramatically in caloric content and macronutrient composition depending on whether it is homemade or sourced from a restaurant. Public nutrition datasets lack consistent mapping from these common dish names to standardized nutrient profiles.

This matters because accurate nutrition tracking is essential for health management, yet current applications impose high user friction. Existing tools require tedious manual ingredient entry or searching through extensive, unverified databases. Users need a system that is fast, consistent, and explainable.

Currently, approaches rely on manual databases (labor-intensive), computer vision (often inaccurate for complex sauces), or ingredient summation (prone to user error). NutriLabelAI addresses this gap by automating the translation of a dish name or image into a complete, standardized FDA nutrition label using modern Natural Language Processing (NLP) and machine learning techniques.

2. Design Objectives

The primary design objective of NutriLabelAI is to create an end-to-end AI system capable of generating FDA-compliant nutrition labels with high accuracy and low user friction. Specifically, the project objectives are:

- (1) Develop a Backend Retrieval System: Create a retrieval engine using FastAPI and PostgreSQL to identify standard dish profiles from a curated database.
- (2) Implement an ML Pipeline: Deploy a machine learning ensemble combining Sentence-BERT embeddings for retrieval, linear regression for scaling, and a neural network for refinement.
- (3) Design a Mobile Frontend: Build a user-friendly React Native (Expo) interface featuring a 4-tab navigation system (Label, History, Explore, Profile).
- (4) Validate System Accuracy: Ensure reliability through evaluation against external datasets and nutritionist expert review.

3. Technical Approach

This section discusses how the team will fulfill the objectives presented above through a structured development pipeline.

3.1. Identifying the Unmet Needs

Through research, we identified that users require results in seconds without manual entry. Current apps require searching through thousands of foods, leading to decision fatigue. There is a specific unmet need for "explainability"—users want to understand why a nutrition estimate was generated. Furthermore, users require the ability to scale a known dish to match their specific calorie targets while maintaining realistic nutrient ratios.

3.2. Determining the Design Constraints

The design is constrained by the following factors:

- **Budget:** The total project cost must remain within the \$200–\$300 range for the duration of development.
- **Platform:** The mobile application must be cross-platform (iOS and Android) compatible, necessitating the use of React Native/Expo.
- **Regulatory:** The output must strictly adhere to FDA Nutrition Facts Label standards (21 CFR 101.36).
- **Latency:** The API inference request must process within acceptable limits for a real-time mobile experience (target <200ms).

3.3. Defining Technical Specifications

The system is based on the modern tech stack, which is designed for scalability:

- **Frontend:** React Native using (Expo) for the mobile interface.
- **Backend:** FastAPI (Python) for handling asynchronous API requests and model inference.
- **Database:** PostgreSQL 14+ with the pgvector extension to handle 384-dimensional vector

embeddings.

- **Machine Learning:** An ensemble model consisting of Sentence-BERT (MiniLM-L6-v2) for retrieval, Linear Regression for scaling, and a Neural Network for final refinement.

3.4. Enumerating Design Concepts

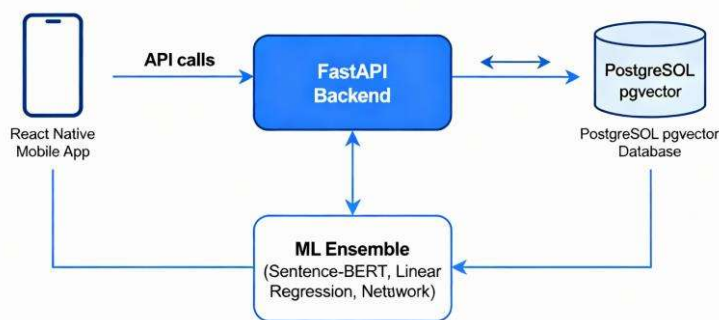
We considered three main design-related concepts:

- **Concept A: Linear Scaling (Text-Only):** The user inputs a text name. The system retrieves the nearest match and uses simple linear algebra to scale nutrients based on the calorie target.
- **Concept B: Multi-Stage Ensemble (Text-Only):** A more advanced approach where text retrieval is followed by a linear regression model, and then refined by a neural network to capture non-linear nutrient relationships (e.g., fat-to-protein ratios in different cuisines).
- **Concept C: Image + Text Hybrid:** The user uploads an image of the food. A computer vision model classifies the dish, which is then fed into the text-based retrieval and scaling pipeline described in Concept B.

3.5. Selecting Design Concept

We selected **Concept C (Image + Text Hybrid)** utilizing the Multi-Stage Ensemble backend. This approach was chosen because it offers the lowest user friction (taking a photo) while maintaining the high accuracy of the ensemble backend. Ensemble aggregation ensures that predictions are grounded in real database values while being flexible enough to scale to user targets.

NutriLabelAI App Clean Architecture Diagram



3.6. Standard Compliance

The project will strictly comply with the **FDA Nutrition Facts Label standard** (21 CFR 101.36) for all visual outputs. Regarding data privacy, user query history will be handled securely, and the system will comply with GDPR/CCPA guidelines where applicable for user data handling. The mobile UI will adhere to WCAG accessibility standards to ensure usability.

for all populations.

4. Project Management

4.1. Team Qualification

- **Yuvaraj Vagula (Team Lead):** Machine Learning Engineering (NLP and Regression models).
- **Nhat Le:** Backend Engineering and ML Infrastructure. Database implementation and development.
- **Rached Arda:** A Computer Science student with a 3.7 GPA and practical experience in system configuration and software maintenance. His technical background includes proficiency in SQL, Python, React Native and Git, which are critical for managing the project.
- **Matthew Lam:** Mobile Frontend Development (React Native/UI).

4.2. Task Assignment

The project tasks are divided between the frontend and backend teams. Yuvaraj and Nhat are responsible for the entire backend infrastructure, including the FastAPI development, database management, and machine learning model training and integration. Rached and Matthew lead the frontend development, specifically focusing on the React Native mobile application, user interface design, and integrating the client-side logic with the backend API.

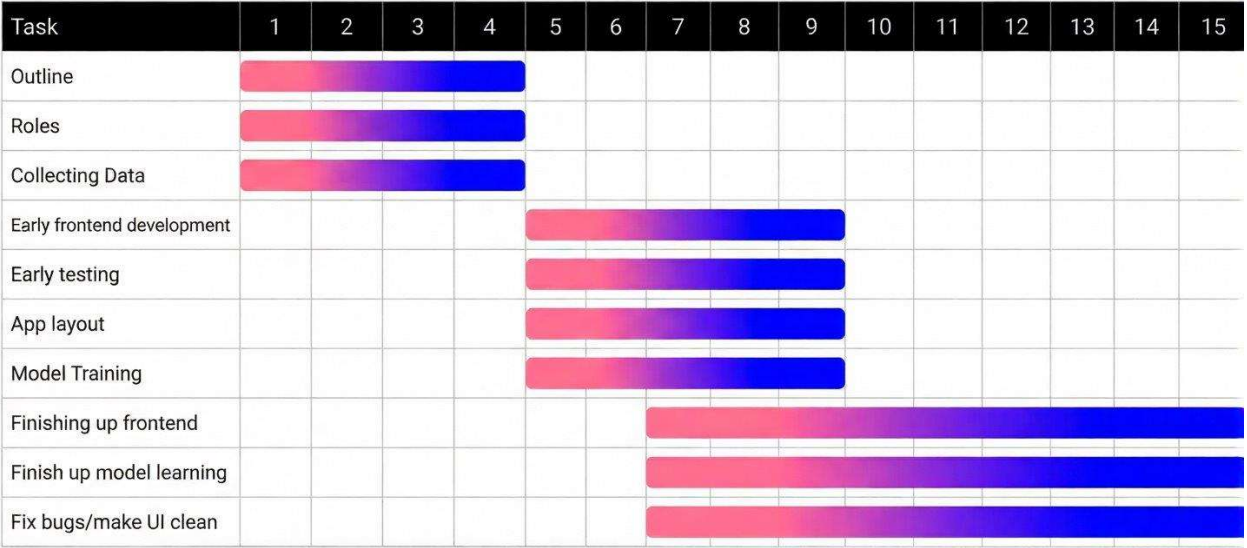
4.3. Timeline

Table 1: Project Milestone Timeline

Phase	Duration	Tasks	Assigned To
Phase 1	Weeks 1-3	Data Collection & Database Setup	Yuvaraj(50%), Nhat(50%)
Phase 2	Weeks 4-6	Model Training & Early Frontend Dev	Rached(50%), Matthew(50%)
Phase 3	Weeks 7-9	API Implementation & App Layout	Rached(50%), Matthew(50%)
Phase 4	Weeks 10-12	Integration of ML into Backend	Yuvaraj(50%), Nhat(50%)
Phase 5	Weeks 13-15	Testing, Validation,	All Team

		and UI Polish	
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Project Gantt Chart



4.4. Deliverables

The team shall deliver:

- Mobile App:** A fully functional iOS/Android app with Image Upload and Results screens.
- API & Backend:** A deployed FastAPI instance with endpoints for /predict and /upload.
- ML Models:** Trained Sentence-BERT and Regression models.
- Deliverables:** A final technical report and GitHub repository.

4.5. Budget

The estimated budget for the project is approximately \$200–\$300.

Table 2: Estimated Project Budget

Item	Supplier	Cost Estimate
API Hosting	DigitalOcean / AWS	\$10–25 / month
Database Hosting	AWS RDS / DigitalOcean	\$12–15 / month
Storage	AWS S3	\$5–10 total

(Embeddings/Models)		
Total Estimated Cost		\$200–\$300

4.6. Communication and Coordination with Sponsor

We will hold weekly status meetings with Dr. Stefan Andrei and submit bi-weekly progress reports via email. A shared GitHub repository will be used for code versioning and document sharing.

5. Professional Awareness

The team recognizes the ethical implications of AI in health. NutriLabelAI provides *estimates*, not clinical advice, and this limitation will be clearly communicated. We are committed to:

- **Ethical AI:** Provide transparency in how the predictions are made.
- **Data Privacy:** Protect user images and search history.
- **Reliability:** Validation of models to prevent misleading health information.

6. References

- [1] U.S. Department of Agriculture, Agricultural Research Service. FoodData Central, 2021.
- [2] Reimers, N., & Gurevych, I. "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," arXiv:1908.10084, 2019.
- [3] U.S. Food and Drug Administration. "Nutrition Labeling of Dietary Supplements," 21 CFR 101.36.
- [4] PostgreSQL Documentation: "pgvector Extension for Vector Similarity Search."
- [5] FastAPI Documentation: "Modern, Fast Web Framework for Building APIs."

Appendix A: Resumes of Team Members

Yuvaraj Vagula:

Yuvaraj Vagula

yuvaraj.vagula.research@gmail.com | 440-413-1079 | linkedin.com/in/raj-vagula | Painesville, OH

Education

Cleveland State University – Honors BS Computer Science, Honors BS Data Science | GPA: 3.61 Aug 2022 – May 2025

- Honors College Scholar | IEEE Board Member | ACM Member

Experience

Applied AI/ML Engineer Intern – Market Leader Technologies, Baltimore, MD Aug 2025 – Present

- Translate cutting-edge research papers into production ML models and backend AI systems.
- Design end-to-end ML pipelines (data preprocessing, training, evaluation) with focus on scalability and client deployment.
- Operationalize theoretical methods from academic research for real-world application and monetization.

ML Research Intern (NSF REU) – UC Riverside, CA Summer 2025

- Invented FORCE, a novel machine unlearning algorithm combining gradient-ascent forgetting with orthogonal subspace projection; achieved 12% higher retention accuracy vs. baselines while maintaining forgetting guarantees.
- Built reproducible PyTorch pipelines with HIPC experimentation and membership inference attack evaluations.
- Co-authoring publication with Prof. Papalexakis and Prof. Tsotras (target: NeurIPS/ICML workshop).

ML & Quantum Computing Research Intern – Cleveland Clinic, OH Dec 2024 – Present

- Developed quantum-classical ML pipelines (Qiskit, PennyLane) for molecular docking optimization in collaboration with IBM Quantum; benchmarked QAOA vs. VQE under simulated noise.
- Automated resource estimation pipeline extracting qubit counts, gate complexity, and circuit depth across 15+ ligands.
- Contributing to grant proposals and manuscripts with clinical and quantum research teams.

Undergraduate Researcher – Cleveland State University Jan 2023 – Present

- Built denoising diffusion models (DDPM, DDIM) on DIV2K dataset; proposed architectural improvements for robustness. Paper under review at Springer Nature Computer Science.
- Investigated ML vulnerabilities under clock-glitch fault injection attacks against AES; paper targeting IEEE.

Lead Engineering Peer Teacher – Cleveland State University Aug 2023 – Present

- Led DSA instruction for 30+ students/semester; improved average exam performance by 15%.
- Trained team of peer teachers; developed curriculum resources in collaboration with faculty.

Publications

- **Unlearning on Tabular Data via Nullspace Projection.** Vagula Y., Blevins A., Wild M., Chen J., Papalexakis E., Tsotras V. *(In Preparation)*
- **Diffusion Models and Techniques for Improved Visual Fidelity.** Vagula Y., Singh A. *Under Review, Springer Nature CS.*
- **Clock Glitch Fault Injection Attacks against AES.** Vagula Y., Mal-Sarkar S. *(Targeting IEEE)*
- **Quantum Computing for Molecular Docking Optimization.** Vagula Y. *(In Preparation)*
- **Interface-Layer Protocols for AI Agents: A Survey of NLWeb and AG-UI.** Vagula Y., Singh A., Chatta N., Ehtesham A. *(In Preparation)*

Skills

Languages: Python, C, Java, SQL **ML/DL:** PyTorch, TensorFlow, Scikit-learn, HuggingFace
Research: Machine Unlearning, Diffusion Models, Membership Inference, Quantum ML (Qiskit, PennyLane)
Tools: Git, Linux, HIPC/Slurm, Jupyter, Docker

Rached Arda:

Rached Arda

9608 Garfield Blvd, Cleveland, OH • 2163551588 • rashedalard4@gmail.com

Computer Science student (GPA 3.7) with foundational development skills in Java, C#, and Python. Eager to apply strong problem-solving and troubleshooting skills. Experienced in identifying system defects and maintaining software on multiple operating systems, including windows/Linux/UNIX environments.

SKILLS

Soft Skills: Communication, Eagerness to adapt new technologies, In-Person Tutoring, Problem-Solving Skills

Languages: C# (C Sharp), HTML, Java, Python

Tools & Databases: Git, GitHub, Microsoft Office, SQL, VS Code

Algorithms & Concepts: Data Structures, Dijkstra's Algorithm, Object-Oriented Programming (OOP)

WORK EXPERIENCE

Sweet Distributing 05/2023 - Present
Sales Associate and Delivery Coordinator Cleveland

Managed the full order lifecycle, including sales input, processing, and final delivery, demonstrating strong attention to detail and process oversight.

SAMA IT 05/2022 - 08/2022
IT Support • Full-time Amman, Jordan

System Configuration and Deployment: Collaborated with the IT support team on the installation and configuration of Microsoft Windows and the Microsoft Office Suite on multiple client and internal systems.

Tutor Cleveland
Personal

Helped my classmates in understanding class materials with different subjects Math, Physics and programming classes

EDUCATION

Bachelor in Computer Science
Cleveland State University • GPA: 3.7 Cleveland, OH

- Community College Transfer Scholarship Recipient
- Choose Ohio First Scholarship Recipient
- Ohio College Opportunity Grant Scholarship Recipient

Associate of Science in Computer Science
Cuyahoga Community College Cleveland, OH

- Awarded summa cum laude
- Mandel Scholarship Recipient

PROJECTS

Route Optimization & Mapping System

Group Project

Engineered and developed the interactive web-based map interface using JavaScript and HTML, serving as the front-end for the route optimization system. Successfully integrated complex Java-based algorithms (Dijkstra's Algorithm, BFS, and DFS) into the front-end map, enabling users to efficiently calculate and visualize the shortest path between dynamic points.



Nhat Thanh Le:

Nhat Thanh Le

Phone: (216) 466-5753 | Email: thanhnhhat465@gmail.com | Location: Cleveland, Ohio

EDUCATION

Cleveland State University

Cleveland, Ohio

Bachelor of Science in Computer Science

- Recipient of Cleveland State University Global Scholarship.
- GPA: 3.0/4.0

EXPERIENCE

Engineering Peer Teacher (EPT): Intro to Programming (Java)

July 2025 – Present

- Lead engaging recitations for 29 students, providing guidance on Java programming concepts including object-oriented programming, data structures, and algorithms.
- Assigned as a Teaching Assistant on lab days to help students with hands-on coding problems.
- Enhanced leadership skills by conducting weekly recitations and improving presentation delivery to engage students.

Engineering Peer Teacher (EPT): Engineering Economy

January 2025 – July 2025

- Assist students in understanding core concepts of engineering economy, including cost analysis, financial decision-making, and economic feasibility studies.
- Collaborate with faculty to identify student needs and tailor learning strategies for diverse groups.
- Mentor peers to enhance their analytical skills and application of engineering economic principles in real-world scenarios.

Math tutoring Per-Calculus, Calculus 1, 2, and 3

June 2024 – September 2024

- Assisted students in improving their grades, test scores, and overall understanding of key mathematical concepts.
- Provided personalized tutoring to students focusing on Pre-Calculus, Calculus 1, 2 and portions of Calculus 3.
- Developed customized lesson plans and strategies to cater to individual learning styles and academic needs.

Teacher's Assistant: My UC English Center in Vietnam

November 2020 – June 2021

- Assisted in teaching basic English to children aged 3-16 at an English center, contributing to their language development and communication skills.
- Collaborated with the professor to create engaging lesson plans and materials for classes of 20 students each semester.

Appendix B: Sponsor Specification

(Not Applicable / Faculty Advised)